Hierarchical Locomotion Control Stack for Quadruped Robots

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Overview

The control architecture maps desired body velocity and heading commands (from operator input or higher-level navigation planners) to motor torques across all joints. The architecture comprises a high-level model predictive controller, a low-level whole-body impulse controller, a dual-stage state estimator, and model-based contact detection.

High-Level Control: NMPC

The upper layer computes optimal Ground Reaction Forces (GRFs) at stance feet to track desired Center of Mass (CoM) trajectories and maintain trunk attitude.

Lumped-Mass Dynamics

The robot is modeled as a lumped point mass subject to gravity and contact forces:

$$m\ddot{\mathbf{p}} = \sum_{i=1}^{n_c} \mathbf{f}_i - m\mathbf{g} \tag{1}$$

$$\frac{d}{dt}(\mathbf{I}\boldsymbol{\omega}) = \sum_{i=1}^{n_c} \mathbf{r}_i \times \mathbf{f}_i \tag{2}$$

where $\mathbf{p} \in \mathbb{R}^3$ denotes CoM position, $\mathbf{f}_i \in \mathbb{R}^3$ the contact force at foot i, m body mass, n_c active stance feet, $\mathbf{g} = [0, 0, g]^\top$ gravitational acceleration, $\mathbf{I} \in \mathbb{R}^{3\times3}$ rotational inertia tensor, $\boldsymbol{\omega}$ angular velocity, and $\mathbf{r}_i = \mathbf{p}_i - \mathbf{p}$ the foot position vector relative to the CoM.

Model Simplifications

To formulate a fast quadratic program (QP) per solve step:

1. **Small Roll and Pitch Angles:** $\dot{\boldsymbol{\Theta}} \approx \mathbf{R}_z(\psi)\boldsymbol{\omega}$ and $\mathbf{I}_G \approx \mathbf{R}_z(\psi)\mathbf{I}_B\mathbf{R}_z(\psi)^\top$, where $\boldsymbol{\Theta} = [\phi, \theta, \psi]^\top$.
2. **Linearization along Nominal Trajectory:** Dynamics are evaluated along commanded yaw angles $\psi(t)$, and foot vectors $\mathbf{r}_i$ are set via scheduled touchdown coordinates.
3. **Decoupled Inertial Cross-Terms:** With low angular rates, $\frac{d}{dt}(\mathbf{I}\boldsymbol{\omega}) \approx \mathbf{I}\dot{\boldsymbol{\omega}}$.

Discrete Optimal Control Formulation

With state $\mathbf{x} = [\boldsymbol{\Theta}^\top, \mathbf{p}^\top, \boldsymbol{\omega}^\top, \dot{\mathbf{p}}^\top]^\top \in \mathbb{R}^{12}$ and force inputs $\mathbf{f} = [\mathbf{f}_1^\top, \ldots, \mathbf{f}_{n_c}^\top]^\top$, the discrete tracking optimization is:

$$\min_{\mathbf{x},\mathbf{f}} \sum_{k=0}^{h-1} \left(\|\mathbf{x}_{k+1} - \mathbf{x}_{k+1}^{\text{ref}}\|_{\mathbf{Q}}^2 + \|\mathbf{f}_k\|_{\mathbf{R}}^2 \right) \tag{3}$$

subject to:

$$\mathbf{x}_{k+1} = \mathbf{A}_k\mathbf{x}_k + \mathbf{B}_k\mathbf{f}_k + \mathbf{g}_k \tag{4}$$

$$|f_{i,x}| \le \mu f_{i,z}, \quad |f_{i,y}| \le \mu f_{i,z}, \quad f_{i,z} > 0 \tag{5}$$

where:

$$\mathbf{A}_k = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_3 & \mathbf{R}_z(\psi_k)\Delta t & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{I}_3 & \mathbf{0}_3 & \mathbf{I}_3\Delta t \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix}, \quad \mathbf{g}_k = \begin{bmatrix} \mathbf{0}_3 \\ \mathbf{0}_3 \\ \mathbf{0}_3 \\ -\mathbf{g}\Delta t \end{bmatrix} \tag{6}$$

$$\mathbf{B}_k = \begin{bmatrix} \mathbf{0}_3 & \cdots & \mathbf{0}_3 \\ \mathbf{0}_3 & \cdots & \mathbf{0}_3 \\ \mathbf{I}_G^{-1}[\mathbf{r}_1\times]\Delta t & \cdots & \mathbf{I}_G^{-1}[\mathbf{r}_{n_c}\times]\Delta t \\ \frac{\mathbf{I}_3\Delta t}{m} & \cdots & \frac{\mathbf{I}_3\Delta t}{m} \end{bmatrix} \tag{7}$$

Decision variables for non-contact swing legs are removed to minimize computational footprint.

Footstep Planning

Touchdown targets are calculated using Raibert's heuristic augmented with centrifugal velocity compensation:

$$\mathbf{r}_i^{\text{cmd}} = \mathbf{p}_{\text{shoulder},i} + \mathbf{p}_{\text{sym}} + \mathbf{p}_{\text{cent}} \tag{8}$$

$$\mathbf{p}_{\text{shoulder},i} = \mathbf{p}_k + \mathbf{R}_z(\psi)\mathbf{l}_i \tag{9}$$

$$\mathbf{p}_{\text{sym}} = \frac{t_{\text{stance}}}{2}\mathbf{v} + k(\mathbf{v} - \mathbf{v}^{\text{cmd}}) \tag{10}$$

$$\mathbf{p}_{\text{cent}} = \frac{1}{2}\sqrt{\frac{h}{g}}\,(\mathbf{v} \times \boldsymbol{\omega}^{\text{cmd}}) \tag{11}$$

where $\mathbf{l}_i$ is nominal hip offset, t_{stance} stance duration, and $k = 0.03$.

Low-Level Control: WBIC

The lower layer resolves joint accelerations and motor torques using Whole-Body Impulse Control (WBIC), tracking MPC forces while stabilizing base posture and swing trajectories.

Rigid-Body Dynamics

$$\mathbf{A}\begin{bmatrix} \ddot{\mathbf{q}}_f \\ \ddot{\mathbf{q}}_j \end{bmatrix} + \mathbf{b} + \mathbf{g} = \begin{bmatrix} \mathbf{0}_{6\times1} \\ \boldsymbol{\tau} \end{bmatrix} + \mathbf{J}_c^\top \mathbf{f}_r \tag{12}$$

where $\mathbf{A}$ is the generalized mass matrix, $\mathbf{b}$ Coriolis forces, $\mathbf{g}$ gravity, $\ddot{\mathbf{q}}_f \in \mathbb{R}^6$ floating-base acceleration, $\ddot{\mathbf{q}}_j \in \mathbb{R}^{n_j}$ joint accelerations, $\boldsymbol{\tau}$ joint torques, $\mathbf{J}_c$ contact Jacobian, and $\mathbf{f}_r$ reaction forces.

Prioritized Task Execution

For prioritized task execution without coupling distortion, tasks are projected onto successive null spaces. For $\mathbf{q} = [\mathbf{q}_f^\top, \mathbf{q}_j^\top]^\top$:

$$\Delta\mathbf{q}_i = \Delta\mathbf{q}_{i-1} + \mathbf{J}_{i|\text{pre}}^\dagger(\mathbf{e}_i - \mathbf{J}_i\Delta\mathbf{q}_{i-1}) \tag{13}$$

$$\dot{\mathbf{q}}_i^{\text{cmd}} = \dot{\mathbf{q}}_{i-1}^{\text{cmd}} + \mathbf{J}_{i|\text{pre}}^\dagger(\dot{\mathbf{x}}_i^{\text{des}} - \mathbf{J}_i\dot{\mathbf{q}}_{i-1}^{\text{cmd}}) \tag{14}$$

$$\ddot{\mathbf{q}}_i^{\text{cmd}} = \ddot{\mathbf{q}}_{i-1}^{\text{cmd}} + \mathbf{J}_{i|\text{pre}}^{\text{dyn}\,\dagger}(\ddot{\mathbf{x}}_i^{\text{cmd}} - \mathbf{J}_i\ddot{\mathbf{q}}_{i-1}^{\text{cmd}} - \dot{\mathbf{J}}_i\dot{\mathbf{q}}) \tag{15}$$

with project intermediate operators defined as:

$$\mathbf{J}_{i|\text{pre}} = \mathbf{J}_i \mathbf{N}_{i-1}, \quad \mathbf{N}_{i-1} = \mathbf{N}_0 \mathbf{N}_{1|0} \cdots \mathbf{N}_{i-1|i-2} \tag{16}$$

$$\mathbf{N}_0 = \mathbf{I} - \mathbf{J}_c^\dagger \mathbf{J}_c, \quad \mathbf{N}_{i|i-1} = \mathbf{I} - \mathbf{J}_{i|i-1}^\dagger \mathbf{J}_{i|i-1} \tag{17}$$

$$\mathbf{J}^{\text{dyn}\,\dagger} = \mathbf{A}^{-1}\mathbf{J}^\top (\mathbf{J}\mathbf{A}^{-1}\mathbf{J}^\top)^{-1} \tag{18}$$

Initialization conditions are $\Delta\mathbf{q}_0 = \mathbf{0}$, $\dot{\mathbf{q}}_0^{\text{cmd}} = \mathbf{0}$, and $\ddot{\mathbf{q}}_0^{\text{cmd}} = \mathbf{J}_c^{\text{dyn}\,\dagger}(-\dot{\mathbf{J}}_c\dot{\mathbf{q}})$. Task errors and acceleration commands follow:

$$\mathbf{e}_i = \mathbf{x}_i - \mathbf{x}_i^{\text{des}} \tag{19}$$

$$\ddot{\mathbf{x}}_i^{\text{cmd}} = \ddot{\mathbf{x}}_i^{\text{des}} + \mathbf{K}_p(\mathbf{x}_i^{\text{des}} - \mathbf{x}_i) + \mathbf{K}_d(\dot{\mathbf{x}}_i^{\text{des}} - \dot{\mathbf{x}}_i) \tag{20}$$

Desired joint positions and velocities feed local PD controllers:

$$\mathbf{q}_j^{\text{cmd}} = \mathbf{q}_j + \Delta\mathbf{q}_j, \quad \dot{\mathbf{q}}_j^{\text{cmd}} = \dot{\mathbf{q}}_j^{\text{cmd}} \tag{21}$$

QP Torque Formulation

Force tracking and floating-base dynamics are optimized via:

$$\min_{\boldsymbol{\delta}\mathbf{f}_r, \boldsymbol{\delta}\ddot{\mathbf{q}}_f} \quad \boldsymbol{\delta}\mathbf{f}_r^\top \mathbf{Q}_1 \boldsymbol{\delta}\mathbf{f}_r + \boldsymbol{\delta}\ddot{\mathbf{q}}_f^\top \mathbf{Q}_2 \boldsymbol{\delta}\ddot{\mathbf{q}}_f \tag{22}$$

$$\text{s.t.} \quad \mathbf{S}_f(\mathbf{A}\ddot{\mathbf{q}} + \mathbf{b} + \mathbf{g}) = \mathbf{S}_f \mathbf{J}_c^\top \mathbf{f}_r \tag{23}$$

$$\ddot{\mathbf{q}} = \ddot{\mathbf{q}}^{\text{cmd}} + \begin{bmatrix} \boldsymbol{\delta}\ddot{\mathbf{q}}_f \\ \mathbf{0}_{n_j} \end{bmatrix} \tag{24}$$

$$\mathbf{f}_r = \mathbf{f}_r^{\text{NMPC}} + \boldsymbol{\delta}\mathbf{f}_r, \quad \mathbf{W}\mathbf{f}_r \geq \mathbf{0} \tag{25}$$

where $\mathbf{S}_f = [\mathbf{I}_6, \mathbf{0}_{6\times n_j}]$ selects the unactuated floating base, $\mathbf{W}$ models friction pyramid bounds, and $\boldsymbol{\delta}\ddot{\mathbf{q}}_f$ relaxes floating-base constraints.

Joint motor torques $\boldsymbol{\tau}_j$ are extracted directly:

$$\begin{bmatrix} \boldsymbol{\tau}_f \\ \boldsymbol{\tau}_j \end{bmatrix} = \mathbf{A}\ddot{\mathbf{q}} + \mathbf{b} + \mathbf{g} - \mathbf{J}_c^\top \mathbf{f}_r \tag{26}$$

State Estimation

A decoupled structure separates nonlinear attitude dynamics from linear translational kinematics.

Attitude Estimation

State vector: $\mathbf{x}_{\text{att}} = [\mathbf{q}_b^{n\top}, \mathbf{b}_\omega^\top]^\top$, with quaternion $\mathbf{q}_b^n \in \mathbb{R}^4$ and gyro bias $\mathbf{b}_\omega \in \mathbb{R}^3$:

$$\dot{\mathbf{q}}_b^n = \frac{1}{2}\begin{bmatrix} 0 & -(\boldsymbol{\omega}_b - \mathbf{b}_\omega)^\top \\ \boldsymbol{\omega}_b - \mathbf{b}_\omega & -\mathbf{S}(\boldsymbol{\omega}_b - \mathbf{b}_\omega) \end{bmatrix} \mathbf{q}_b^n, \quad \dot{\mathbf{b}}_\omega = \mathbf{0} \tag{27}$$

To avoid magnetic interference from high motor currents, a constant pseudo-North reference vector is mapped from the navigation frame. The exogenous Kalman filter (ExKF) linearizes about stable nonlinear observer estimates $\hat{\mathbf{x}}$:

$$\dot{\hat{\mathbf{x}}}_{\text{att}} = \mathbf{f}(\hat{\mathbf{x}}) + \mathbf{F}(\hat{\mathbf{x}}_{\text{att}} - \hat{\mathbf{x}}) + \mathbf{K}\left(\mathbf{z} - \mathbf{h}(\hat{\mathbf{x}}) - \mathbf{H}(\hat{\mathbf{x}}_{\text{att}} - \hat{\mathbf{x}})\right) \tag{28}$$

$$\dot{\mathbf{P}} = \mathbf{F}\mathbf{P} + \mathbf{P}\mathbf{F}^\top - \mathbf{K}\mathbf{H}\mathbf{P} + \mathbf{Q}, \quad \mathbf{K} = \mathbf{P}\mathbf{H}^\top \mathbf{R}^{-1} \tag{29}$$

Translational Kinematics

State vector: $\mathbf{x} = [\mathbf{p}_n^\top, \mathbf{v}_n^\top]^\top \in \mathbb{R}^6$ with input $\mathbf{u} = \mathbf{R}_b^n \mathbf{f}_{\text{ib}}^b - \mathbf{g}_n$:

$$\dot{\mathbf{x}} = \begin{bmatrix} \mathbf{v}_n \\ \mathbf{u} \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \mathbf{0}_3 & \mathbf{I}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 \end{bmatrix} \tag{30}$$

Leg odometry fuses kinematic foot velocities over n_s stance contacts:

$$\dot{\mathbf{x}}_l^b = -\alpha_l \mathbf{J}_l \dot{\mathbf{q}}_l - \boldsymbol{\omega}_b \times \mathbf{x}_l^b, \quad \dot{\mathbf{x}}^b = \frac{1}{n_s}\sum_{l=1}^{n_s} \dot{\mathbf{x}}_l^b \tag{31}$$

$$\mathbf{z} = \mathbf{R}_b^n \dot{\mathbf{x}}^b, \quad \mathbf{H} = \begin{bmatrix} \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix} \tag{32}$$

Covariance $\mathbf{R}$ inflates during detected slip events.

Contact Detection

Contact schedules $\alpha_i \in \{0, 1\}$ gate leg odometry and MPC stance horizons.

Generalized Momentum Observer

Using $\mathbf{p} = \mathbf{M}(\mathbf{q})\dot{\mathbf{q}}$, external torque $\boldsymbol{\tau}_{\text{ext}} = \sum \mathbf{J}_{ci}^\top \mathbf{F}_{\text{ext},i}$ is estimated without requiring joint acceleration signals:

$$\dot{\mathbf{p}} = \mathbf{C}^\top(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} - \mathbf{G}(\mathbf{q}) + \boldsymbol{\tau} + \mathbf{J}^\top \mathbf{F}_{\text{ext}} \tag{33}$$

To reject velocity quantization noise, a linear Kalman filter models force dynamics:

$$\begin{bmatrix} \dot{\mathbf{p}}_j \\ \dot{\mathbf{F}}_{\text{ext},j} \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{J}_j^\top \\ \mathbf{0} & \mathbf{A}_F \end{bmatrix} \begin{bmatrix} \mathbf{p}_j \\ \mathbf{F}_{\text{ext},j} \end{bmatrix} + \begin{bmatrix} \mathbf{I} \\ \mathbf{0} \end{bmatrix} \mathbf{u} + \boldsymbol{\omega} \tag{34}$$

IMM-KF Hybrid Estimation

Foot-ground interaction represents a switched dynamical system. An Interacting Multiple-Model Kalman Filter (IMM-KF) executes parallel observers across flight, stance, and impact models. A pseudo-force measurement assists mode discrimination:

$$\mathbf{F}_{\text{pf}} = -(\mathbf{J}\mathbf{M}^{-1}\mathbf{J}^\top)^\dagger \left(\mathbf{J}\mathbf{M}^{-1}(\boldsymbol{\tau} - \mathbf{C}\dot{\mathbf{q}} - \mathbf{G}) + \dot{\mathbf{J}}\dot{\mathbf{q}}\right) \tag{35}$$

Contact states α_i correspond to the maximum-probability mode.

Identification & Sim-to-Real

Parameter identification applies variable-frequency chirp trajectory excitations to the ungrounded robot to estimate link inertias, CoM offsets, and joint friction. Kinematic and dynamic descriptions exported via URDF are validated in closed-loop simulations within MuJoCo and Isaac Sim, incorporating sensor noise and actuator delays prior to hardware deployment.